

Social Robot Intelligence Quotient to Measure Human-Robot Interaction Performance for Human-Care Service Robots

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Abstract In order to measure human-robot interaction (HRI) performance for human-care service robots, a concept of social robot intelligence quotient (SRIQ) is proposed, which is composed of five element intelligence quotients (EIQs): Human Recognition, User Modeling, Body Movement, Self-Expression, and Communication. The SRIQ is assessed as the weighted sum of the EIQs where the weight factors are determined according to the morphology of the robot subject. Each EIQ is also assessed as the weighted sum of the scores for its component skill factors which actually measure their performance. Preliminary evaluation tests are conducted for the SRIQ elements with a robot subject called Jenny and the test results verify the feasibility of the SRIQ as a system-level evaluation measure for the social robot intelligence.

I. INTRODUCTION

Earlier robots have contributed a lot to the manufacturing industry for several decades with their strong, agile and repetitive actuation performance in structured environments. More recently, as robot sensory and computational capabilities have improved even in ill-structured environments, robots have gradually pervaded human daily-living society providing cognitive human-care services [1][2][3]. In order for robots to be suitably integrated into human society and to provide effective human-care services, they require a certain level of social intelligence to work with people and each other [4].

Social intelligence required for human-care service robots is unlikely to cover a wide variety of human society features but desired to attain minimal interaction-capability level, i.e. to exhibit the following human social characteristics: to express/perceive emotions; to communicate with high-level dialogue; to learn/recognize models of other agents; to establish/maintain social relationships; to use natural cues (gaze, gestures, etc.); to exhibit distinctive personality and character; to learn/develop social competencies [5]. In this respect, there have been a lot of efforts so far to develop socially interactive robots with social intelligence approaching to the animal or human level in part. It has been widely accepted by roboticists that these robots are called as 'social robot' in general. Recent social robots such as AIBO (Sony), Buddy (Blue Flog Robotics), Jibo (Jibo Inc.), etc., with advances in artificial intelligence and robotics technologies, have become to attain quite high levels of social intelligence interacting with humans and environments [6][7].

However, these socially interactive robots providing human-care services have not been so successful. One of the reasons why social robot technologies have not succeeded in revitalizing practical robot market is that useful measures or tools to evaluate social robot intelligence in the system level have hardly appeared yet. In this paper, we come up with a new concept of Social Robot Intelligence Quotient(SRIQ) as a measure to evaluate the system-level intelligence in HRI, and a scoring method to assess the SRIQ as well as its Element Intelligence Quotients(EIQs) is proposed.

This paper is organized as follows. In Section 2, we break down the social robots into four classes with respect to morphology, select the five key EIQs constituting the SRIQ, and propose a method to assess the SRIQs by summing up their scores in accordance to the morphological class that the robot subject belongs to. Section 3 describes a detailed method to score the EIQs which practically evaluate how intelligent the robot subject is in the sense of social HRI. In section 4, preliminary evaluation tests of EIQs are conducted to instantiate the SRIQ, using a robot platform based on Pepper robot. Finally, we make concluding remarks in section V.

II. SOCIAL ROBOT INTELLIGENCE QUOTIENT(SRIQ)

As for social robots, the expected interaction capabilities are usually influenced by their morphology and thus the social robot intelligence has to be assessed differently according to the morphological class where the social robot subject belongs to. Considering the dependency of social robot intelligence on the morphology, we classify social robots into 4 broad categories: anthropomorphic, zoomorphic, caricatured and functional, as shown in Fig. 1 [5].

Generally speaking, social robots in any morphological class providing human-care services operate as partners, peers or assistants, which means that they need to exhibit a certain degree of adaptability and flexibility to drive the interaction with a wide range of humans. In order for social robots to be accepted as natural interaction partners, what sorts of social interaction skills are essentially required?

Here, we start by the definition of robot to extract requirements for social interaction skills. According to the ISO8273:2012 robotics vocabulary standard [8], robot was defined as "actuated mechanism programmable in two or more axes with a degree of autonomy, moving within its

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environment, to perform intended tasks". A social robot, as a kind of robot, is typically equipped with the actuated or moving mechanism so as to express their intention or to move the body within the interaction distance zones [9], where a degree of autonomy describes the level of skills in its expression and motion. The intended tasks for social robots to perform are also deeply related to the social interaction with human inside the interaction distance, and therefore social HRI skills in human recognition, body movement and self-expression/communication are prerequisite for achieving the intended task goals.

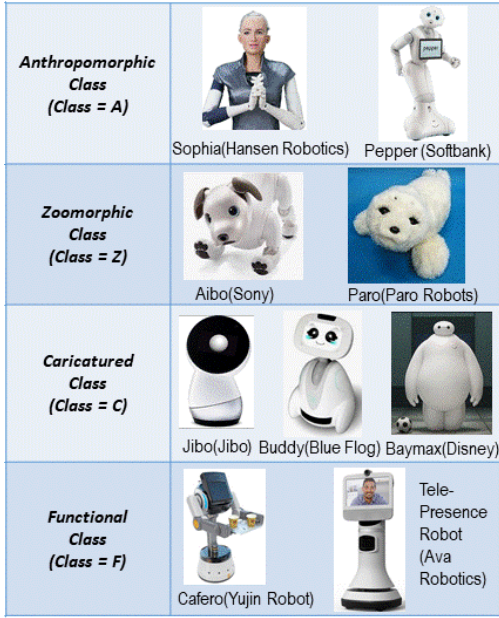


Figure 1. Morphological classes of social robots[5]

In this paper, taking into account the morphological classification and prerequisite requirements for HRI skills, we propose to quantify the social robot intelligence quotient (SRIQ) as the weighted sum of the following five key Element Intelligence quotient(EIQ)s, which imply levels of the essential HRI skills for the social robot subjects to exhibit in general: Human Recognition (HR); User Modeling (UM); Body Movement (BM); Self-Expression (SE); Communication (CO), as shown in TABLE I.

Every EIQ is basically weighted in equal, i.e. $W_{Class}(id) = 0.2$, but the EIQ weights may be selectively assigned as decimals between 0.1 and 0.3, the total sum of which should be 1.0. Referring to the basic equal weight, the selective weight of each EIQ is assigned in the way that a decimal value is added or subtracted as follows: +0.05 if a corresponding intelligence is more important, +0.10 if it is very important, -0.05 if it is less important, and -0.10 if it is least important, as shown in TABLE II. The skill factors determining EIQs will be explained later in details with the evaluation methods. Considering the importance of EIQs for each morphological class, we propose the EIQ weights $\{W(HR), W(UM), W(BM), W(SE), W(CO)\}$ as follows: $\{0.20, 0.15, 0.20, 0.20, 0.25\}$ for Class A; $\{0.20, 0.20, 0.15, 0.25, 0.20\}$ for Class Z; $\{0.20, 0.20, 0.10, 0.25, 0.25\}$ for Class C; $\{0.25, 0.10, 0.30, 0.20, 0.15\}$ for Class F.

For instance, as for a humanoid robot, when the weighting factors are assigned as $\{W(HR), W(UM), W(BM), W(SE), W(CO)\} = \{0.20, 0.15, 0.20, 0.20, 0.25\}$ and also the EIQs are evaluated as $EIQ(HR) = 80$, $EIQ(UM) = 60$, $EIQ(BM) = 70$, $EIQ(SE) = 50$ and $EIQ(CO) = 70$, the social robot intelligence quotient of the robot subject becomes $SRIQ-A = 66.5$.

TABLE I. THE SRIQ ASSESSMENT WITH 5 ESSENTIAL EIQS

EIQ id	EIQ Name	Descriptions (Element Skill Factors)	EIQ weight W_{Class} [%]			
			Class = A	Class = Z	Class = C	Class = F
HR	Human Recognition	Face Recognition, Human Detection/Tracking, Gender/Age Recognition, Behavior Recognition	$W_A(HR)$	$W_Z(HR)$	$W_C(HR)$	$W_F(HR)$
UM	User Modeling	Extrinsic/Intrinsic Attributes, User Belongings, User Intention	$W_A(UM)$	$W_Z(UM)$	$W_C(UM)$	$W_F(UM)$
BM	Body Movement	Approaching to User, Human-Avoiding /User-Accompanying Navigation, User Following, Passing Object, Handshake	$W_A(BM)$	$W_Z(BM)$	$W_C(BM)$	$W_F(BM)$
SE	Self-Expression	Expressional motion, Emotion Control Scheme, Voice Synthesis, Personality	$W_A(SE)$	$W_Z(SE)$	$W_C(SE)$	$W_F(SE)$
CO	Communication	Verbal/Non-verbal dialog, Interaction stance, Behavior Learning	$W_A(CO)$	$W_Z(CO)$	$W_C(CO)$	$W_F(CO)$

Given $EIQ(id)$ s.t. $0 \leq EIQ(id) \leq 100$ for $id \in \{HR, UM, BM, SE, CO\}$
 $SRIQ-Class = \sum_{id} W_{Class}(id) \cdot EIQ(id)$ for $W_{Class}(id)$ s.t. $\sum_{id} W_{Class}(id) = 1.0$
 where Class $\in \{A, Z, C, F\}$

TABLE II. AN EXAMPLE TO DETERMINE THE EIQ WEIGHTS (CLASS A)

HR		UM		BM		SE		CO	
Face Recognition	O	Cloth Feature	X	Human Avoidance	Δ	Expressional Motion	Δ	Verbal Dialog	O
Gender/Age Recognition	Δ	Hair Feature	X	Human Following	Δ	Emotion Control Model	Δ	Interaction Stance	O
Motion/Gesture Recognition	Δ	User Belongings	Δ	Human Guiding Nav.	Δ	Voice Synthesis	Δ	Non-Verbal Dialog	O
Human Tracking	Δ			Object Passing	Δ	Personality	X	Behavior Learning	Δ
				Handshaking	Δ				
Normal		Less Important		Normal		Normal		More Important	
W(HR)=0.20		W(UM)=0.15		W(BM)=0.20		W(HR)=0.20		W(HR)=0.25	

(Note) Importance of Skill Factors is symbolized as O(Important), Δ (normal), X(Not Important)

III. A METHOD TO SCORE THE ELEMENT INTELLIGENCE QUOTIENTS(EIQS)

A. Human Perception (HR)

In order for a social robot to provide human-care services, it is basically important for the robot to recognize not only the specific person who requires the service but also what he or she is doing now in quite complicated environments. This human recognition capability is thus selected as the first EIQ evaluation item which consists of four skill factors: 1) Face Recognition, 2) Gender/Age Recognition; 3) Motion Gesture Recognition; and 4) Human Tracking. The Human Recognition EIQ is also evaluated as the weighted sum of the five skill factor scores as shown in TABLE III.

TABLE III. EVALUATION OF THE HUMAN ROGNITION EIQ

Code	Skill factor	Descriptions
HR ₁	Face Recognition	face identification for the registered persons
HR ₂	Gender/Age Recognition	recognition of gender/age of the human partner
HR ₃	Motion/Gesture Recognition	recognition of specified motion gestures in interaction distance
HR ₄	Human Tracking	Detection/tracking of the human partner moving around

$EIQ(HR) = \sum_i W_{Class}(HR_i) \cdot \text{Score}(HR_i)$
 where $\sum_i W_{Class}(HR_i) = 1.0$ for $i = 1, 2, 3, 4$ and $0 \leq \text{Score}(HR_i) \leq 100$

The recognition intelligence for three skill factors except the human-tracking mainly relies on the recognition rate and the response time in general. Especially, the response time becomes important when the recognition results are

immediately used for HRI in task-executing environments. In HCI study, it is well known that a system should not take more than two seconds after input to response [10]. In HRI, robot usually recognizes human or environment, makes decisions and takes actions, and thus the recognition response time lower than two seconds should be assured before decisions and actions are made. In this respect, we propose a method to score the recognition intelligence of the three skill factors ($Code = HR_i$ for $i = 1, 2, 3$) as the following equation:

$$Score(Code) = W_{RT} * Recog(Code) \quad (1)$$

where $Recog(Code)$ [%] is the recognition rate and W_{RT} is the weighting factor for the response time such that

$$W_{RT} = \begin{cases} 1 & \text{if } RT \leq 0.5 \text{ sec} \\ 1.5 - RT & \text{if } 0.5 \leq RT \leq 1.5 \text{ sec} \\ 0 & \text{if } RT \geq 1.5 \text{ sec} \end{cases} \quad (2)$$

The face recognition in computer vision has become one of the hottest topics in innovative AI and deep learning technologies and its algorithms are applicable to robot vision for social robots without change. The evaluation of the face recognition performance typically needs big datasets for advanced deep-learning, and many kinds of database for face recognition have been announced and tested so far such as Multi-PIE, IJB-A, etc. In this paper the Labeled Faces in the Wild (LFW) database [11], one of the most popular face DBs, is adopted to evaluate the recognition rate of the face recognition skill factor.

The representative intrinsic human attributes, gender and age, are quite useful information for providing user-friendly robot services to the human partner and so the gender/age recognition is selected as an important skill factor of Human Recognition EIQ. The intelligence of the gender/age recognition skill factor is evaluated through a practical test for the robot to identify gender and age of the human partner in front of the robot camera. The recognition rate $Recog(HR_2)$ is calculated as follows:

$$1) Recog(HR_2) = (average\ gender\ recognition\ rate + average\ age\ recognition\ rate)/2,$$

$$2) average\ gender\ recognition\ rate = \frac{1}{N} \sum_{n=1}^N R_n^g$$

where $gender\ recognition\ rate R_n^g$ is 100[%] if it is succeeded and 0 if it is failed, and N is # of subjects,

$$3) average\ age\ recognition\ rate = \frac{1}{N} \sum_{n=1}^N R_n^a$$

where $age\ recognition\ rate R_n^a$ is calculated by

$$R_n^a[\%] = \begin{cases} 100 & \text{if } |Ea| \leq 5 \\ 10 \times (15 - |Ea|) & \text{if } 5 \leq |Ea| \leq 15 \\ 0 & \text{if } |Ea| \geq 15 \end{cases} \quad (3)$$

for Ea = the actual age – the recognized age.

The age recognition error Ea has ± 5 year tolerance margin, but the recognition has no meaning with the error beyond ± 15 year and the error in between two ends makes the recognition rate linearly decrease, as in (3).

As for the motion/gesture, the skill factor intelligence is evaluated through a practical test for the robot to identify 10 gesture commands from the human partner inside 1.5 m ~ 3.5 m with a RGB-D camera, as proposed in [12]. The evaluation test is conducted as follows: an evaluation database is built with 10 kinds of 1,000 gesture datasets for 20 subjects acquired

from the RGB-D camera on the robot; The recognition algorithm is run, randomly selecting the learning datasets and the test datasets from the database; the average recognition rate and the response time are calculated after 10 repetitive tests.

The human-tracking performance is evaluated by use of a test database with video clips recorded by an RGB-D camera mounted on a mobile robot with 1 m height. The video-recording environment is set up as follows: the operating area is larger than 10m x 10m, the background looks natural composed of a sofa and bright solid color wallpaper, human subjects are sitting down on the sofa or moving with natural speed, and two tables are randomly set to show the occlusion situation. The database includes 512x424 depth image with skeleton information of 25 joints [13].

B. User Modeling (UM)

User models generally contain attributes that describe a user or a group of users, and so help the robot not merely understand human behavior/dialogue but adapt the robot's behavior to accommodate users with varying knowledge and skills [5]. The User Modeling EIQ is proposed as the second evaluation item, which consists of three skill factors: 1) Cloth Feature Recognition, 2) Hair Feature Recognition, 3) User-Belonging Recognition, as shown in TABLE IV.

TABLE IV. EVALUATION OF THE USER MODELLING EIQ

Code	Skill factor	Descriptions
UM ₁	Cloth Feature Recognition	recognition of cloth attributes such as color, type, length, etc.
UM ₂	Hair Feature Recognition	recognition of hair attributes such as color, length, style, etc.
UM ₃	User-Belonging Recognition	recognition of user-belonging attributes such as glass, phone, etc.
$EIQ(UM) = \sum_i W_{class}(UM_i) * Score(UM_i) / 100$ where $\sum_i W_{class}(UM_i) = 1.0$ for $i = 1, 2, 3$ and $0 \leq Score(UM_i) \leq 100$		

The User Modeling EIQ is also evaluated as the weighted sum of the four skill factor scores, each of which is assessed as in (1) and (2) with $Code = UM_i$ for $i = 1, 2, 3$. The recognition rate $Recog(Code)$ is calculated as in the following equation:

For the required number of attributes L , the actual number of attributes $M(\leq L)$ and the number of human subjects N

$$Recog(UM_i)[\%] = \frac{1}{L} \sum_{m=1}^M (\frac{1}{N} \sum_{n=1}^N R_n^m) \text{ for } i=1,2,3 \quad (4)$$

where R_n^m is the recognition rate of the m^{th} attributes for the n^{th} human subject.

The skill factor of cloth recognition is evaluated with 4 attributes among color/sleeve-length/gender-type/style/pattern of clothes, and that of hair recognition is done with 2 attributes among color/length/style of hairs, for more than 10 human subjects. The skill factor of user belonging recognition is assessed with 4 attributes among hat, purse, pill box, glasses and socks for more than 2 human subjects [14].

C. Body Movement (BM)

Social robots have outstanding values in providing human-care services compared to the other devices since they can move their bodies in interaction with human users intimately. The Body Movement EIQ is therefore selected as one of EIQ evaluation components, which consists of the five skill factors: 1) Human-Avoiding Navigation; 2) Human-Following; 3) Human-Guiding Navigation, 4) Object Passing; and 5) Hand-Shaking, as shown in TABLE V.

The intelligence of the five skill factors ($Code = BM_i$ for $i = 1, 2, 3, 4, 5$) is evaluated via real-world tests and their scores are assessed by the methods as in TABLE VI.

TABLE V. EVALUATION OF THE BODY MOVEMENT EIQ

Code	Skill factor	Descriptions
BM ₁	Human-Avoiding Navigation	navigating to the target while avoiding the oncoming person
BM ₂	Human Following	following the human partner going ahead while maintaining the interaction distance
BM ₃	Human-Guiding Navigation	guiding the humans along the given paths while maintaining the interaction distance
BM ₄	Object Passing	passing objects to the human partner without dropping
BM ₅	Hand-shaking	shaking hands with the human partner
$EIQ(BM) = \sum_i W_{class}(BM_i) \cdot Score(BM_i) / 100$ $\text{where } \sum_i W_{class}(BM_i) = 1.0 \text{ for } i = 1, 2, 3, 4, 5 \text{ and } 0 \leq Score(BM_i) \leq 100$		

TABLE VI. SCORE ASSESSMENT METHOD FOR THE BM SKILL FACTORS

Code	Score(BM _i) [%]	Real-World Test Scenario
BM ₁	(# of success for the robot to navigate from the initial position to the final while avoiding the approaching human in the opposite direction) / (# of tests = 20)	- Initial position: the robot is opposite to the human with 10 m distance in a 2.5 m wide hallway - Final position: the robot and the human are located at the opposite side from the initiation - The average moving speeds of both robot and human are more than 0.5 m/sec - Tolerance error bound of location: ± 0.5 m
BM ₂	(# of success for the robot to follow the target human inside 1.5 ~ 3.5m distance for 2 min) / (# of tests = 20)	- Indoor test area: wider more than 100 m ² - # of human moving around including the target user: more than 3 persons - All the other environments are the same as human-tracking ($Code = HR_4$)
BM ₃	(# of success for the robot to navigate from the initial position to the final via the stopover positions in 2 min) / (# of tests = 20)	- Indoor test area: width ≥ 5 m and length ≥ 20 m - 5 stopover positions within ± 0.5 m error for 5 sec - # of human to guide: more than 3 persons - each person maintains the distance to the robot in 3.5 m
BM ₄	(# of success for the robot to receive the book from the user and pass the cup to the user in 30 sec) / (# of tests = 20)	- Test scenario: the user passes a sample book to the robot; the robot recognizes the position of the book and receives it; the robot put the book on the nearby table; the robot grasps and passes a sample cup to the user; the robot releases it after recognizing that the user grasped it.
BM ₅	(# of success for the robot to extend the hand in 5 sec and shake hands in 10 sec) / (# of tests = 20)	- Test scenario: the user extends the right hand to the robot; the robot recognizes the hand position and also extends the robot hand to the position; the robot grasps the user hand after confirming the user's grasping; the robot shakes the hand 3 times upside down

D. Self-Expression (SE)

Social robots typically have the ability to express their own internal states such as emotions, body posture or personality by means of gesture, facial expression, voice, etc. This self-expression capability is selected as a key EIQ evaluation item which is composed of the following four skill factors:

- Expressional Motion: Robots usually express their internal states via motions such as facial expression (with eyes, eyebrows, or lip motion), emotional gesture, etc.
- Emotion Model: Robots can express themselves referring to their own emotion model such as pre-specified emotional patterns, affective loop generation, long-term interaction, etc.
- Voice Synthesis: Various types of voices are excellent means for self-expression such as recorded voice, customized Text-To-Speech(TTS), pitch/tempo- controlled TTS [15], etc.
- Personality: If user adjust the robot personality according to the preference, the robot expressions are predictable and so it will enhance the interaction skills. Here, Big-5 personality factors (extraversion, agreeableness, dependability, emotional stability, openness) are taken into account [16].

The Self-Expression EIQ is evaluated as the weighted sum of the four skill factor scores, as shown in TABLE VII. Each skill factor is scored by the following equation,

$$Score(Code) = Level * Quality(Code) / N\text{-level} \quad (5)$$

where *Level* stands for the intelligence level with higher integer value for higher skill, *N-level* is the number of intelligence levels, and *Quality(Code)* [%] denotes the degree of quality in the corresponding intelligence level, as for the skill factor represented by *Code* in TABLE VII.

TABLE VII. EVALUATION OF THE SELF-EXPRESSION EIQ

Code	Skill factor	Descriptions
SE ₁	Expressional Motions	facial expression (with eyes, eyebrows, or lip motion), emotional gesture, etc.
SE ₂	Emotion Control Model	pre-specified emotional patterns, affective loop generation, long-term emotion control, etc.
SE ₃	Voice Synthesis	recorded voice, Text-To-Speech (TTS), pitch/tempo control in TTS, etc.
SE ₄	Personality	the big-five personality factors
$EIQ(SE) = \sum_i W_{class}(SE_i) \cdot Score(SE_i)$ $\text{where } \sum_i W_{class}(SE_i) = 1.0 \text{ for } i = 1, 2, 3, 4 \text{ and } 0 \leq Score(SE_i) \leq 100$		

With respect to the expressional motion ($Code=SE_1$), the intelligence of the skill factor is broken down into the following four levels: (level 1) More than two internal states can be expressed via a screen display; (Level 2) More than five internal states (e.g. five emotions: happiness, anger, fear, surprise and sadness) can be expressed via a screen display; (Level 3) More than two internal states can be expressed via actuation of facial mechanisms or gestures; (Level 4) More than five internal states can be expressed via actuation of facial mechanisms or gestures.

Quality(SE₁) [%] represents the percentage of completeness evaluated by the following procedure: 1) the evaluation administrator decides what level the robot subject is able to achieve; 2) the prepared test scenario corresponding to the level is practically performed by the robot subject in front of the evaluator group consisting of more than 5 robot experts; 3) the evaluator group respond to the questionnaire with the following semantic difference scale [17] as in Fig. 2; 4) the mean value of the scores for all evaluators is calculated. For instance, if the 5 evaluators score the average values for the level 3 scenario as {3, 4, 3, 5, 4}, then the mean value is 3.8 and *Quality(SE₁)* becomes 76%. Therefore, *Score(SE₁)* is calculated to be 57 in (5), since *N-Level* = 4 and *Level* = 3.

Completeness									
(1 st time)	Low	1	2	3	4	5	High	----	()
(2 nd time)	Low	1	2	3	4	5	High	----	()
(3 rd time)	Low	1	2	3	4	5	High	----	()
(4 th time)	Low	1	2	3	4	5	High	----	()
(5 th time)	Low	1	2	3	4	5	High	----	()
Average Value : ()									

Figure 2. A sample questionnaire for an expert evaluator

The skill factor intelligence of the emotion control model ($Code=SE_2$) forms a hierarchy with the following four levels: (Level 1) Several pre-specified patterns of emotional expressions are carried out without emotion model; (Level 2) Several pre-specified patterns of emotional expressions are executed according to the recognized short-term emotions of user; (Level 3) User's emotion detection by robot, emotional

behavior generation and emotion elicitation in user form a so-called ‘*Affective Loop*’ [18]; (Level 4) Emotions are learnt and expressed based on a long-term emotion control model with an affective loop. Quality(SE₂) [%] and Score(SE₂) are scored as in the evaluation process of the expressional motion skill factor(SE₁).

The intelligence levels of the speech synthesis skill factor (Code=SE₃) are described as follows: (Level 1) The recorded voices are played according to the pre-specified situation; (Level 2) The TTS(Text-To-Speech) voices with a constant tone are played according to the pre-specified situation; (Level 3) The TTS voices with the pre-assigned parameters such as pitch and tempo according to the emotional states are played in the corresponding state; (Level 4) The TTS voices, whose parameters are adjusted according to the current emotional state, are played. Quality(SE₃) [%] and Score(SE₃) are also scored as in the evaluation process of the expressional motion skill factor(SE₁).

The intelligence of the personality skill factor (Code=SE₄) is evaluated with five levels, each of which represents the implemented number of the orthogonal ‘big-five’ personalities [16]. The scoring method of Quality(SE₄) [%] is the same as that of Quality(SE₁) [%], but N-level is 5 to calculate Score(SE₄) in (5).

E. Communication (CO)

Natural human-robot communication is an essential functional element for socially interactive robots to successfully carry out the intended tasks. The communication capability is thus selected as the second EIQ evaluation item which is composed of the four skill factors: 1) Verbal Dialog, 2) Interaction Stance, 3) Non-verbal Dialog, and 4) Behavior Learning, as shown in TABLE VIII. The Communication EIQ is also evaluated as the weighted sum of the four skill factor scores, and each skill factor is assessed via the same process to evaluate the skill factors of Self-Expression EIQ as in (5).

TABLE VIII. EVALUATION OF THE COMMUNICATION EIQ

Code	Skill factor	Descriptions
CO ₁	Verbal Dialog	Verbal dialog communication considering service domains, speaker dependency/independency, natural language processing, etc.
CO ₂	Interaction Stance	Capability of entering/maintaining interaction distances, turning/looking toward human partner, etc.
CO ₃	Non-verbal Dialog	Communication via gestures, gazing, turn-taking, etc.
CO ₄	Behavior Learning	behavior model rendering, behavior mimicry, context awareness, etc.
$EIQ(CO) = \sum_i W_{class}(CO_i) * Score(CO_i)$ $\text{where } \sum_i W_{class}(CO_i) = 1.0 \text{ for } i = 1, 2, 3, 4 \text{ and } 0 \leq Score(CO_i) \leq 100$		

The intelligence of the verbal dialog skill factor is classified into the following five levels: (Level 1) The robot subject carries out every elementary command in a pre-defined one-word command set with more than 10 elements; (Level 2) The robot subject carries out every elementary command in a pre-defined one-sentence voice command set with more than 10 elements; (Level 3) The robot subject understands the meaning of more than 10 speaker-dependent voice sentence and responds to it in a specific service domain; (Level 4) The robot subject understands the meaning of the speaker-independent voice sentence and responds to it in a specific service domain; (Level 5) The robot subject communicates with human partner via natural language processing(NLP) in all service domains.

The intelligence hierarchy of the interaction stance skill factor is represented by the following four levels: (Level 1) The robot subject can recognize the direction of the human partner and only turn its body towards the partner; (Level 2) The robot subject can recognize the direction of the human partner’s eyes and move to look the eyes; (Level 3) The robot can approach to the human partner inside the social distance zone [10] while looking the eyes of the partner; (Level 4) The robot subject can naturally turn its body and face toward the partner while keeping the social distance (1.2 m ~ 3.5 m).

As for the non-verbal dialog skill factor, four intelligence levels are considered as follows: (Level 1) Pre-defined body motions or gestures of the robot subject, the activation data of which is tagged on the words or sentences to be spoken, are taken in interaction; (Level 2) The robot subject maintains eye gazing with the partner in interaction; (Level 3) The robot subject takes body motions autonomously, aware of the context of the words or sentences that the partner speaks in interaction; (Level 4) The robot subject takes its turn and resume dialog with gesture, aware of the situation in dialog.

The behavior learning skill factor is proposed to have the following four intelligence levels: (Level 1) The robot subject takes motions according to the pre-defined behavior model; (Level 2) The primitive behaviors mimicking those of human are learnt and implemented; (Level 3) Goal-achieving composite behaviors mimicking those of human are learnt and implemented in performing intended tasks; (Level 4) A situation-motivation-behavior model is learnt and the robot subject takes motions aware of the context based on the model.

IV. PRELIMINARY SRIQ EVALUATION TESTS

In order to verify the feasibility of the SRIQ evaluation method, we have conducted preliminary evaluation tests for the SRIQ elements, i.e. the five EIQs, in a bottom-up manner. The social robot we used for evaluation is named ‘Jenny’ with the physical body as shown in Fig.1. Jenny was designed for providing elderly-care services to the aged who live alone at home. The working environment of Jenny is a kind of living lab, consisting of a living room, a kitchen, a bedroom, a balcony and a bathroom. Jenny is composed of a Pepper robot body made by Softbank Robotics, a RGB-D camera for vision, a LiDAR sensor module for navigation. Jenny is thus capable of interacting with human via voice and vision while navigating around the home environment. Evaluation tests have been conducted so far partially for the five SRIQ elements with the Jenny robot and some preliminary test results are shown in TABLE IX.

As for the human recognition EIQ, four component skill factors scored 91.81 for face recognition (HR₁), 92.7 for gender/age recognition (HR₂), 46.63 for motion/gesture recognition (HR₃) and 95 for human tracking (HR₄), thru tests in the living lab. The EIQ(HR) thus scored 81.54 with even weight value 0.25 for each component skill factor. For the user modeling, EIQ(UM) became 54.13 with even weight value 1/3 since three component skill factors scored 85.33, 0 and 77.05. EIQ(BM), EIQ(SE) and EIQ(CO) scored 20.0, 22.5 and 46.0, respectively, based on the evaluation scores of each component skill factors as shown in TABLE IX with even weight value. In particular, the robot has reasonable communication capability with level4 verbal and non-verbal

dialog skills, level3 interaction stance skill, even though it cannot learn behaviors. Qualities of 4 component skill factors of Communication (CO) were evaluated 80% for all and thus their scores became {64, 60, 60, 0}. Therefore, the overall social robot intelligence quotient for the reference humanoid robot ‘Jenny’, SRIQ-A, finally scored 44.43 when we assigned weighting values of Class A as in TABLE II to the Element Intelligence Quotients.

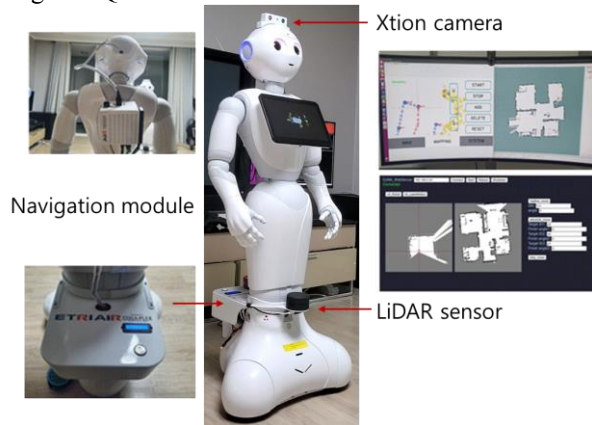


Fig. 1. A reference social robot platform ‘Jenny’

TABLE IX. PRELIMINARY TEST RESULTS FOR THE SRIQ ELEMENTS

EIQ	Code	Score [%]	Notes
Human Recognition	HR ₁	91.81	Frontal face considering pose variation [19]
	HR ₂	92.7	Age recognition: 94.5, gender recognition: 90.9 [20]
	HR ₃	46.63	Self-test in the given environment [12]
	HR ₄	95	Self-test in the given environment [13]
User Modeling	UM ₁	85.33	4 attributes: color, length, style and pattern of clothes
	UM ₂	-	Not implemented yet
	UM ₃	77.05	Top 5 attributes: hat, purse, pill box, glasses, socks
Body Movement	BM ₁	-	Not implemented yet
	BM ₂	-	Not implemented yet
	BM ₃	-	Not implemented yet
	BM ₄	-	Not implemented yet
	BM ₅	100	Handshaking motions are successful always
Self-Expression	SE ₁	60	Level 3 with high quality is achieved by self-test
	SE ₂	30	Level 2 with normal quality is achieved by self-test
	SE ₃	-	Not implemented yet
	SE ₄	-	Not implemented yet
Communication	CO ₁	64	Level 4 with high quality is achieved by self-test
	CO ₂	60	Level 3 with high quality is achieved by self-test
	CO ₃	60	Level 4 with high quality is achieved by self-test
	CO ₄	0	Not implemented yet

V. CONCLUDING REMARKS

Quite a lot of social robots to provide human-care services have appeared into the real world with social intelligence, but useful measures or tools to evaluate social robot intelligence in the system level have hardly appeared yet. To address this issue, we proposed a new concept of Social Robot Intelligence Quotient(SRIQ) as a measure to evaluate the system-level intelligence in human-robot interaction(HRI), and also methods to score the SRIQ and the five Element Intelligence Quotient(EIQ)s which constitute it.

The EIQs were scored through the real-world evaluation tests by repetitive experiments and/or by human expert evaluators, but preliminary evaluation tests have been conducted partially so far. In further studies, the evaluation methods should be enhanced in the direction where human

expert evaluators are minimally involved and all the skill factor items are tested. The proposed evaluation framework can be used as a cornerstone for raising the intelligence of various social robots and resultantly improving the practical quality of robot services provided to human.

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